

Ahad Khan - Computational Chemistry Projects

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I did three small projects to learn computational chemistry on my own, using the same five food antioxidants (quercetin, catechin, ascorbic acid, caffeic acid, rutin) throughout so the projects connect to each other. Each one looks at the molecules in a different way: first in vacuum, then in moving water, then in still water.

Project 1: Gas-Phase DFT

WHAT I WAS TRYING TO FIND OUT

Which of these five antioxidants is the most reactive, and why, based purely on its electron structure (no solvent involved)?

WHAT I USED

ORCA for the DFT calculations (B3LYP/6-31G*), Multiwfn to look at the HOMO/LUMO orbitals, ChimeraX to view the electron density maps.

WHAT I FOUND

Quercetin had the smallest HOMO-LUMO gap (3.97 eV) of the five, which lines up with it being known as a strong antioxidant. The electron-rich spots on its structure matched where antioxidants are known to donate electrons from.

github.com/ahadk210-cmyk/Ahad_Khan_Computational_Chemistry

Project 2: Molecular Dynamics

WHAT I WAS TRYING TO FIND OUT

How does a molecule actually behave when it's surrounded by real water molecules over time, instead of sitting still in a calculation?

WHAT I USED

GROMACS for the simulation, GROMOS54A7 as the force field, ran a 500 ps simulation of ethanol in water, checked stability with RMSD.

WHAT I FOUND

The system stayed stable the whole 500 ps (RMSD of 0.361 Å), so the simulation worked properly end to end - setup, equilibration, and the actual run.

github.com/ahadk210-cmyk/MD_Project

Project 3: Solvation Effects (CPCM)

WHAT I WAS TRYING TO FIND OUT

Does the answer from Project 1 change once you put the same five molecules in water instead of vacuum?

WHAT I USED

Same ORCA setup as Project 1, but added an implicit water model (CPCM) this time and reran all five molecules.

WHAT I FOUND

Quercetin and caffeic acid got easier to react with in water (their gap got smaller). Rutin did the opposite - its gap got bigger by 17%, even though it looks a lot like quercetin. Probably because of the extra sugar group attached to it.

github.com/ahadk210-cmyk/CPCM_Solvation

All the code, input files, and results for each project are on GitHub, with a README explaining what was done and how.